Infosys - Coding 2

A linear sequence of N positions is defined, where each position i (from 1 to N) contains an integer value V[i]. Two robots, A and B, are initially placed at the extreme ends of the sequence. Robot A starts at position 1, and Robot B starts at position N. Upon being placed, both robots immediately collect the values at their respective starting positions. The robots operate in alternating turns, with Robot A always taking the first turn. On its turn, the active robot must choose to move toward the opponent's current position by a distance of either 1 or 2 steps. If a robot chooses to move a distance of 1 to a new cell, it collects the value contained in that cell. If a robot chooses to move a distance of 2, it bypasses the intermediate cell and only collects the value at its final destination. The game concludes immediately when a robot's chosen move would cause it to occupy or pass the current position of its opponent. In this final, game-ending move, the moving robot does not collect any additional points, regardless of the value at the destination. Robot A acts to maximize its own total score. Robot B acts to minimize Robot A's total score. Find the maximum total score Robot A can guarantee for itself, assuming Robot B plays optimally to minimize Robot A's score.

Input:

5
10 2 100 3 20

Output:

110

Explanation: A collects 10 at position 1. On its first move it jumps 2 to position 3, collecting 100. B then plays to block, and A cannot safely collect more. Total = 10 + 100 = 110.

Input Format

First line: integer N, the number of positions. Next N lines: each contains an integer V[i].

Constraints

1 <= N <= 500 -10^9 <= V[i] <= 10^9

Output Format

The output contains an integer value

Sample Input 0

```
5
10
2
100
3
20
```

Sample Output 0

```
110
```

Explanation 0

A collects 10 at position 1. On its first move it jumps 2 to position 3, collecting 100. B then plays to block, and A cannot safely collect more. Total = 10 + 100 = 110.

Sample Input 1

```
1
7
```

Sample Output 1

```
7
```